

# Lecture 2. Probability (Part 2)

## 1 Commonly Used distributions

- Discrete Distributions
- Continuous Distributions
- Sampling Distributions

## 2 Types of Convergence

## 3 Law of Large Number

## 4 Central Limit Theorem

**Reading:** Rice Ch2-3, Ch6; Casella and Berger 3.1-3.5, 4.6, 5.1-5.3, 5.5.

**Further Reading:** Roussas 3.2-3.4, 5.2, 6.3

# Some Commonly Used Distributions

**Question:** For a given random phenomenon or data type, what distribution (or statistical model) is more appropriate to depict it?

Discrete distributions. Corresponding pmfs are usually specified.

(1) **Uniform distribution**  $U(a_1, \dots, a_m)$ : Equal probability to obtain  $a_1, \dots, a_m$ .

- **pmf:**  $p(x) = \begin{cases} \frac{1}{m}, & x = a_1, \dots, a_m \\ 0, & \text{otherwise} \end{cases}$
- **mgf:**  $\frac{\sum_{j=1}^m e^{a_j t}}{m}$
- **mean:**  $\frac{\sum_{j=1}^m a_j}{m} \equiv \bar{a}$
- **variance:**  $\frac{\sum_{j=1}^m (a_j - \bar{a})^2}{m}$
- **parameter:**  $a_i \in R, i = 1, \dots, m$
- **example:** throw a fair die once.

(2) **Bernoulli distribution  $B(p)$**  . A Bernoulli distribution takes on only two values, 0 and 1, with probabilities  $1 - p$  and  $p$ , respectively.

- **pmf**:  $p(x) = \begin{cases} p^x(1-p)^{1-x}, & \text{if } x = 0 \text{ or } x = 1 \\ 0, & \text{otherwise} \end{cases}$
- **mgf**:  $pe^t + 1 - p$
- **mean**:  $p$
- **variance**:  $p(1 - p)$
- **parameter**:  $p \in [0, 1]$
- **example**: toss a coin once,  $p$ =probability that head occurs

**Note.** If  $A$  is an event, then the indicator random variable  $I_A$  follows the Bernoulli distribution.

(3) **Binomial distribution**  $B(n, p)$  . Suppose that  $n$  independent Bernoulli trials are performed, where  $n$  is a fixed number. The total number of 1 appearing in the  $n$  trials follows a binomial distribution with parameters  $n$  and  $p$ .

- **pmf**:  $p(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x}, & x = 0, 1, \dots, n \\ 0, & \text{otherwise} \end{cases}$
- **mgf**:  $(pe^t + 1 - p)^n$
- **mean**:  $np$
- **variance**:  $np(1-p)$
- **parameter**:  $p \in [0, 1]$  and  $n = 1, 2, \dots$
- **example**: number of heads, toss a coin  $n$  times.

**Note.**  $(a+b)^n = \sum_{x=0}^n \binom{n}{x} a^x b^{n-x}$ .

- Most common application in finance: security price modeling.  
Assumed that over the next small interval of time security price will rise (a success) or fall (a failure) by a given amount.

**Example 1.17 (The stock market in a one-period binomial model).** We will assume that on our market at the time point  $t = 0$  there are both of the following investment opportunities:

- Purchase and (short-) selling of stocks with today's price

$P_1(0) = p_1 > 0$  and future price

$$X(T) = \begin{cases} f \cdot e^{rT} + g \cdot p_1 \cdot u, & \text{with probability } p \\ f \cdot e^{rT} + g \cdot p_1 \cdot d, & \text{with probability } (1 - p) \end{cases}$$

where  $u > d$  is true. The pair  $(f, g)$  is a trading strategy,  $f$  is the face amount invested and  $g$  represents the number of stocks kept at  $t = 0$ . If  $f$  is a negative number  $\Rightarrow$  a loan was taken up. If  $g$  is negative  $\Rightarrow$  a short sale has taken place.

- Fixed deposit or loan at an interest rate of  $r \geq 0$ , in which case we adopt the continuous return in a time frame  $[0, T]$ , i.e. the capital development of a monetary unit is given by means of

$$P_0(0) = 1, \quad P_0(T) = e^{rT}.$$

**Explanation:** The investor can distribute his capital at  $t = 0$ , buy or sell stocks, and invest or lend money.

In this model, the number of upward movements of the stock price is  $B(1, p)$ -distributed.

**Exercise 6.** Given the model in Example 1.17 and that opening capital  $x > 0$ , (a) calculate  $E(X(T))$  and  $Var(X(T))$ .  
(b) Suppose that  $r = 0.05, u = 1.2, d = 1, T = 1, p_1 = 100, p = 0.75$ . imagine being in possession of an opening capital of S\$1000. Determine all trading strategies  $(f, g)$  with  $E(X(T)) \geq 1100$ . Which of these has the minimal variance?

**Example 1.18.** The ratio of boys to girls at birth in Singapore is quite high at 1.09 : 1. What proportion of Singapore families with exactly 6 children will have at least 3 boys? (Ignore the prob of multiple births.) [Interesting trivia: In most countries the ratio of boys to girls is about 1.04 : 1, but in China it is 1.15 : 1]

**Exercise 7.** A manufacturer of metal pistons finds that on the average, 12% of his pistons are rejected because they are either oversize or undersize. What is the probability that a batch of 10 pistons will contain (a) no more than 2 rejects? (b) at least 2 rejects?

**(4) Poisson distribution  $P(\lambda)$ .** Limit of binomial distribution

$X_n \sim B(n, p_n)$ , where  $p_n \rightarrow 0$  as  $n \rightarrow \infty$  in such a way that  $\lambda_n \equiv np_n \rightarrow \lambda$ .

$$\begin{aligned}
 & \binom{n}{x} p_n^x (1 - p_n)^{n-x} \\
 &= \frac{n(n-1) \dots (n-x+1)}{x!} \left( \frac{\lambda_n}{n} \right)^x \left( 1 - \frac{\lambda_n}{n} \right)^{n-x} \\
 &= \frac{n(n-1) \dots (n-x+1)}{n^x} \frac{1}{x!} \lambda_n^x \left( 1 - \frac{\lambda_n}{n} \right)^{n-x} \\
 &= 1 \left( 1 - \frac{1}{n} \right) \dots \left( 1 - \frac{x-1}{n} \right) \frac{\lambda_n^x}{x!} \left( 1 - \frac{\lambda_n}{n} \right)^n \left( 1 - \frac{\lambda_n}{n} \right)^{-x} \\
 &\rightarrow 1^x \cdot \frac{\lambda^x}{x!} \cdot e^{-\lambda} \cdot 1 = \frac{\lambda^x e^{-\lambda}}{x!}
 \end{aligned}$$

[Note: if  $a_n \rightarrow a$ ,  $(1 + \frac{a_n}{n})^n \rightarrow e^a$ .]

If  $n$  large, the pmf of  $B(n, p)$  is not easily calculated. Then we can approximate them the pmf of  $P(\lambda)$ , where  $\lambda = np$ .

Handwritten:  $\gamma \sim \text{poisson distribution}$   
 $m_Y(t) = \sqrt{(e^t - 1)}$   
 $\gamma > 0$

Handwritten: Let  $v = e^\lambda \Rightarrow \lambda = \ln v, Y \sim \text{pois}(\lambda)$

- **Explanation.** Let  $X$  be the number of times some event occurs in a given time interval  $I$ . Divide the interval into many small subintervals  $I_k, k = 1, \dots, n$ , of equal length. Let  $N_k$  be the number of events occurring in  $I_k$ . When we can assume  $N_1, \dots, N_n$  are independent and approximately  $\sim B(P)$ ,  $X$  has a distribution near  $P(\lambda)$ , where  $\lambda = np$

Handwritten:  $X \sim P(\lambda)$

- **pmf of  $P(\lambda)$ :**  $p(x) = \begin{cases} \frac{\lambda^x}{x!} e^{-\lambda}, & x = 0, 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$

- **mgf:**  $e^{\lambda(e^t - 1)}, t \in \mathbb{R}$

Handwritten derivation of mgf:  
 $m_X(t) = E(e^{tx}) = \sum_{x=0}^{\infty} e^{tx} \cdot \frac{\lambda^x}{x!} e^{-\lambda} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x e^{tx}}{x!} = e^{-\lambda} \cdot e^{\lambda e^t} = e^{\lambda(e^t - 1)}$

- **mean:**  $\lambda$ , average number of events per unit

- **variance:**  $\lambda$  [Note.  $e^\lambda = \sum_{x=0}^{\infty} \frac{\lambda^x}{x!}$ ]

- **parameter:**  $\lambda > 0$

- **example:** number of phone calls coming into an exchange during a unit of time.

Handwritten:  $M_{\sum_{i=1}^k X_i}(t) = \prod_{i=1}^k m_{X_i}(t) = \prod_{i=1}^k e^{\lambda_i(e^t - 1)} = e^{(\sum_{i=1}^k \lambda_i)(e^t - 1)}$   
 $Y \sim \text{pois}(\sum_{i=1}^k \lambda_i)$

- Let  $X_i \sim P(\lambda_i), i = 1, \dots, k$ , and  $X_1, \dots, X_k$  are independent. Then  $Y = X_1 + \dots + X_k \sim P(\lambda_1 + \dots + \lambda_k)$ .

**Example 1.19.** Find the probability of 4 customers arriving in 3 minutes when the mean is 3.6.

**Exercise 8.** Information which causes market price to move arrive at a rate of 10 pieces per minute. Find the probability of only eight pieces of information arriving in the next minute ?



just have some impression  
no need to memorize pdf----

(5) **Negative Binomial distribution**  $NB(r, p)$ . Suppose that a sequence of independent Bernoulli trials is performed until the appearance of the  $r$ th 1. Let  $X$  be the total number of trials, then  $X$  follows negative binomial distribution.

- **pmf**:  $p(x) = \begin{cases} \binom{x-1}{r-1} p^r (1-p)^{x-r}, & x = r, r+1, \dots \\ 0, & \text{otherwise} \end{cases}$
- **mgf**:  $\frac{p^r e^{rt}}{[1 - (1-p)e^t]^r}, t < -\log(1-p)$
- **mean**:  $\frac{r}{p}$
- **variance**:  $\frac{r(1-p)}{p^2}$
- **parameter**:  $p \in [0, 1], r = 1, 2, \dots$
- **example**: lottery, no. of tickets a person must purchase up to and including the  $r$ th winning ticket.

(6) **Hypergeometric distribution**  $HG(r, n, m)$ . Suppose that an urn contains  $n$  black balls and  $m$  white balls. Let  $X$  denote the number of black balls drawn when taking  $r$  balls without replacement. Then  $X$  follows hypergeometric distribution.

- **pmf**: 
$$p(x) = \begin{cases} \frac{\binom{n}{x} \binom{m}{r-x}}{\binom{n+m}{r}}, & x = 0, 1, \dots, \min(r, n), r-x \leq m \\ 0, & \text{otherwise} \end{cases}$$

[Note:  $\binom{n+m}{r} = \sum_x \binom{n}{x} \binom{m}{r-x}$ .]

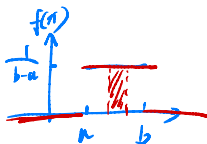
- **mgf**: No explicit form.
- **mean**:  $\frac{rn}{n+m}$  average number of events per unit
- **variance**:  $\frac{rnm(n+m-r)}{(n+m)^2(n+m-1)}$
- **parameter**:  $r, n, m = 1, 2, \dots, r \leq n+m$
- **example**: sampling industrial products for defect inspection

**Note.** Relationship between hypergeometric and binomial distributions: Let  $m, n \rightarrow \infty$  in such a way that  $p_{m,n} \equiv \frac{n}{m+n} \rightarrow p$ , where  $0 < p < 1$ . Then

$$\frac{\binom{n}{x} \binom{m}{r-x}}{\binom{n+m}{r}} \rightarrow \binom{r}{x} p^x (1-p)^{r-x}.$$

## Continuous Distributions

$$U(0,1)$$



$$P(X \in (\pi_i, \pi_{i+1} \Delta \pi)) \\ = P(\pi \in (\pi_i, \pi_{i+1} \Delta \pi))$$

(7) Uniform distribution  $U(a,b)$ . Choose a number at random between  $a$  and  $b$ .

• pdf:  $f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$

• cdf:  $F(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & x > b \end{cases}$

$a=0, b=1, U(0,1) F(x)=x \quad x \in (0,1)$   
 $m_Y(t) = \frac{e^t}{t}$

• mgf:  $\frac{e^{bt}-e^{at}}{t(b-a)}, t \in \mathbb{R}$   $m_X(e^{tx}) = \int_a^b e^{tx} \cdot \frac{1}{b-a} dx =$

$Y \sim U(0,1)$

• mean:  $\frac{a+b}{2}$   $E(Y) = \frac{1}{2}, Y \sim U(0,1)$

• variance:  $\frac{(b-a)^2}{12}$

• parameter:  $a, b \in \mathbb{R}, a < b$

probability integral transformation

Let  $Y \sim F_Y(Y)$   $W = F_Y(Y)$  so  $W \sim U(0,1)$

Note.  $U(0,1)$  is useful for pseudo-random number generation  
 Any CDF  $F(x)$  known, we generate a  $U(0,1)$  to...

$Y = F_Y^{-1}(X), X \sim U(0,1)$

CDF of  $w$ .  $P(W \leq w) = P(F_Y(Y) \leq w) = F_Y(F_Y^{-1}(w)) = P(Y \leq F_Y^{-1}(w)) = F_Y(F_Y^{-1}(w)) = w$

(8) Exponential distribution  $EXP(\lambda)$ .

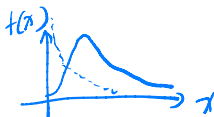
- pdf:  $f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$
- cdf:  $F(x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$
- mgf:  $\frac{\lambda}{\lambda - t}, t < \lambda$
- mean:  $\frac{1}{\lambda}$
- variance:  $\frac{1}{\lambda^2}$
- parameter:  $\lambda > 0$
- example: lifetime or waiting time

$$\begin{aligned}
 T &\sim \text{Exp}(\lambda) & F(t) &= 1 - e^{-\lambda t} \\
 P(T > t_0 + t_1 | T \geq t_0) &= \frac{P(T > t_0 + t_1, T \geq t_0)}{P(T \geq t_0)} = \frac{P(T > t_0 + t_1)}{P(T \geq t_0)} \\
 &= \frac{e^{-\lambda(t_0 + t_1)}}{e^{-\lambda t_0}} = e^{-\lambda t_1} \quad \text{free of } t_0
 \end{aligned}$$

Note.

- Memoryless.
- Relationship between exponential and Poisson distributions. Let  $T_1, T_2, \dots$  be i.i.d.  $\sim \text{EXP}(\lambda)$  and  $S_k = T_1 + \dots + T_k$ ,  $k = 1, 2, \dots$ . Let  $X_i$  be the number of  $S_k$ 's that falls in  $[t_{i-1}, t_i]$ ,  $i = 1, \dots, n$ , then  $X_1, \dots, X_n$  are independent, and  $X_i \sim P(\lambda(t_i - t_{i-1}))$ . The reverse statement is also true.

Question. How to interpret  $\lambda$ ?  $\lambda$  is the incident rate per time unit



### (9) Gamma distribution $\text{Gamma}(\alpha, \lambda)$ .

- pdf:  $f(x) = \begin{cases} \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$   $\alpha=1 \Rightarrow \text{Exp}(\lambda)$
- mgf:  $\left(\frac{\lambda}{\lambda-t}\right)^\alpha, t < \lambda$
- mean:  $\frac{\alpha}{\lambda}$
- variance:  $\frac{\alpha}{\lambda^2}$
- parameter:  $\alpha, \lambda > 0$
- example: used to model the default rate of credit portfolios by CreditRisk

**Note 1.**  $\alpha$ : shape parameter;  $\lambda$ : scale parameter

**Note 2.** gamma function  $\Gamma(\alpha) \equiv \int_0^\infty y^{\alpha-1} e^{-y} dy$ ,  $\Gamma(1) = 1$ ,  
 $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ ,  $\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1)$ .

True  $w$   $w_1, \dots, w_n$   
 $w_i = w + \sum_{k=1}^n \varepsilon_k$

$m(t) = e^{at+bt^2}$   
 $a = \mu, b = \sigma^2/2$

(10) Normal distribution  $N(\mu, \sigma^2)$ . plays a central role in Prob. and Stat. (e.g., CTL)

pdf:  $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, x \in \mathbb{R}$

mgf:  $e^{\mu t + \frac{\sigma^2 t^2}{2}}, t \in \mathbb{R}$

mean:  $\mu$

variance:  $\sigma^2$

parameter:  $\mu \in \mathbb{R}, \sigma > 0$

$N(0,1)$  standard normal distribution

pdf  $f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$   
 $m_X(t) = \int_{-\infty}^{\infty} e^{tx} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$

only by method of pdf or mgf to show it is normal distribution

knowing expectation and variance cannot show its normal distribution

Note.

1 Bell-shaped pdf.

2  $X \sim N(\mu, \sigma^2) \Rightarrow$  for  $a, b \in \mathbb{R}$ ,  $aX + b \sim N(a\mu + b, a^2\sigma^2)$ . In particular,  $\frac{X-\mu}{\sigma} \sim N(0,1)$ .

3 Let  $X_1, \dots, X_k$  be i.i.d.  $\sim N(\mu, \sigma^2)$ . Then  $\bar{X}_k \in N(\mu, \frac{\sigma^2}{k})$

$E(aX+b) = aEX+b = a\mu+b$   
 $Var(aX+b) = a^2 Var(X) = a^2 \sigma^2$   
 $M_{aX+b}(t) = e^{bt} M_X(at) = e^{bt} \cdot e^{\mu at + \frac{\sigma^2 a^2 t^2}{2}} = e^{(a\mu+b)t + \frac{a^2 \sigma^2 t^2}{2}}$

we can show normal distribution of  $aX+b$

$X_{i/k} \sim N(\frac{\mu}{k}, \frac{\sigma^2}{k^2})$

$\bar{X} = \sum_{i=1}^k X_{i/k} / k$ ,  $m_{\bar{X}}(t) = \frac{1}{k} \sum_{i=1}^k m_{X_{i/k}}(t)$

$m_{\bar{X}}(t) = \frac{1}{k} \sum_{i=1}^k e^{\mu t + \frac{\sigma^2 t^2}{2k}} = e^{\mu t + \frac{\sigma^2 t^2}{2k}}$

$\sum_{i=1}^k \mu = k\mu = \mu$ ,  $\sum_{i=1}^k \frac{\sigma^2}{k} = \sigma^2$

$$X_i \sim \text{Normal}(\mu, \sigma^2)$$

# Sampling Distributions $\frac{1}{n} \sum_{i=1}^n X_i$

- Sample statistic:** Let  $X_1, \dots, X_n$  be independently and identically distributed (i.i.d.) random sample. A function of  $X_i$ 's (without unknown parameters) is called a **statistic**. The sampling distribution is the distribution of a statistic.

## (11) Chi-square distribution $\chi_n^2$ .

$$Z_i = \frac{X_i - \mu}{\sigma} \sim N(0, 1)$$

- pdf:**  $f(x) = \begin{cases} \frac{1}{\Gamma(\frac{n}{2})2^{n/2}} x^{\frac{n}{2}-1} e^{-\frac{x}{2}}, & x \geq 0 \\ 0, & x < 0 \end{cases}$

$$\sum_{i=1}^n Z_i^2 \sim \chi_n^2$$

- mgf:**  $\left(\frac{1}{1-2t}\right)^{\frac{n}{2}}$

$$\alpha = \frac{n}{2}, \lambda = \frac{1}{2}$$

- mean:**  $n$

- variance:**  $2n$

- parameter:** degree of freedom  $n = 1, 2, \dots$

$$M_{\sum X_i}(t) = \prod_{i=1}^n M_{X_i}(t) = \prod_{i=1}^n \left(\frac{1}{1-2t}\right)^{\frac{n_i}{2}} = \left(\frac{1}{1-2t}\right)^{\sum \frac{n_i}{2}} = \left(\frac{1}{1-2t}\right)^{\frac{\sum n_i}{2}}$$

$$\sum X_i \sim \chi_{\sum n_i}^2$$

**Note.**

- Special case of the Gamma distribution,  $\chi_n^2 = \text{Gamma}\left(\frac{n}{2}, \frac{1}{2}\right)$

- Let  $X_1, \dots, X_k$  be independent and  $X_i \sim \chi_{n_i}^2$ . Then

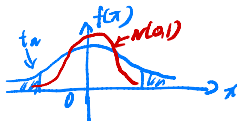
$$Y = X_1 + \dots + X_k \sim \chi_{n_1 + \dots + n_k}^2$$

- Let  $Z \sim N(0, 1)$ , then  $X = Z^2 \sim \chi_1^2$

$$N(0, 1)^2 \sim \chi_1^2$$

(12) t distribution  $t_n$ .  $\frac{N(0,1)}{\sqrt{\chi^2_n/n}} \rightarrow \text{independent} \Rightarrow t_n$

- pdf:  $f(x) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{n\pi}\Gamma(\frac{n}{2})} \left(1 + \frac{t^2}{n}\right)^{-\frac{n+1}{2}}, x \in \mathbb{R}$
- mgf: No explicit form
- mean: 0,  $n > 1$
- variance:  $\frac{n}{n-2}, n > 2$   $\frac{1}{n-2} > 1$   $N(0,1)$   $n \rightarrow \infty$
- parameter: degree of freedom  $n = 1, 2, \dots$



Note.

- Let  $Z \sim N(0,1)$  and  $U \sim \chi_n^2$  be independent, then  $\frac{Z}{\sqrt{U/n}} \sim t_n$
- $f(x) = f(-x)$ , symmetric about 0
- As  $n \rightarrow \infty$ ,  $t_n$  tends to  $N(0,1)$  (by LLN)
- $t_n$  has heavier tail than  $N(0,1)$

(13) F distribution  $F_{m,n}$ .

- pdf: Complicated function form
- mgf: No explicit form
- mean:  $\frac{n}{n-2}, n > 2$
- variance:  $\frac{2n^2(m+n-2)}{m(n-2)^2(n-4)}, n > 4$
- parameter: degrees of freedom  $m, n = 1, 2, \dots$

Note.

① Let  $U \sim \chi_m^2$  and  $V \sim \chi_n^2$  be independent, then  $\frac{U/m}{V/n} \sim F_{m,n}$

② Let  $X \sim t_n$ , then  $Y = X^2 \sim F_{1,n}$

③  $X \sim F_{m,n} \Rightarrow X^{-1} \sim F_{n,m}$

$$F_{m,n} = \frac{1}{F_{n,m}}$$

$$\text{indep} \left( \frac{v/a}{u/n} \sim F_{n,m} \right)$$

$$X \sim \frac{N(0,1)}{\sqrt{\chi_n^2/n}} \sim t_n$$

$$\begin{aligned} \chi^2 &= \frac{M(0,1)}{\chi_n^2/n} \text{ indep} \\ &= \frac{\chi_1^2/1}{\chi_n^2/n} \sim F(1,n) \end{aligned}$$

# Mean and variance of normal random sample

**Distributions of sample mean and sample variance.** Let  $X_1, \dots, X_n$  be i.i.d.  $\sim N(\mu, \sigma^2)$ . Define

- **Sample mean:**  $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$
- **Sample variance:**  $S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$

Then

①  $\bar{X}_n \sim N(\mu, \frac{\sigma^2}{n})$  and  $\sqrt{n}(\bar{X}_n - \mu)/\sigma \sim N(0, 1)$ .

②  $\bar{X}_n$  and  $S_n^2$  are independent.

③  $(n-1)S_n^2/\sigma^2 \sim \chi_{n-1}^2$

④  $\frac{\sqrt{n}(\bar{X}_n - \mu)}{S_n} \sim t_{n-1}$ .

standardize:  $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = a\bar{X} + b$

$H_0: \sigma^2 = \sigma_0^2$ , Stat:  $\frac{(n-1)S_n^2}{\sigma_0^2} \sim \chi_{n-1}^2$

$H_0: \mu = \mu_0$

under  $H_0$

**Question:** Are  $(X_1 - \bar{X}_n, \dots, X_n - \bar{X}_n)$  independent?

Why the degree of freedom =  $n - 1$ , rather than  $n$ ?

Sample variance  $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$   $x_1 - \bar{x}, \dots, x_n - \bar{x}$

In fact  $\sum_{i=1}^n (x_i - \bar{x}) = n\bar{x} - n\bar{x} = 0 \Rightarrow \sum_{i=2}^n (x_i - \bar{x}) = - (x_1 - \bar{x})$

$$\therefore s^2 = f^* \left( \underbrace{x_1 - \bar{x}}_{y_1}, \dots, \underbrace{x_n - \bar{x}}_{y_n} \right) \quad (x_1, \dots, x_n) \quad (y_1, \dots, y_n)$$

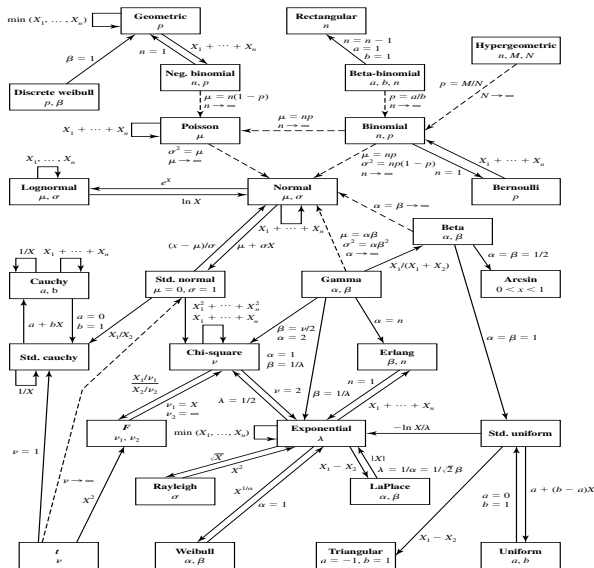
$$\text{Let } \left. \begin{array}{l} y_1 = x_1 - \bar{x} \\ y_2 = x_2 - \bar{x} \\ \vdots \\ y_n = x_n - \bar{x} \end{array} \right\}$$

## \*Some Other Distributions.

- **Cauchy distribution**  $C(\mu, \sigma)$ :  $f(x) = \frac{\sigma}{\pi} \frac{1}{\sigma^2 + (x - \mu)^2}, x \in \mathbb{R}$ .  
Heavy-tailed distribution, applied to govern asset returns
- **Log-normal distribution**:  
 $\log(X) \sim N(\mu, \sigma^2) \Rightarrow X \sim \text{lognormal}(\mu, \sigma^2), X > 0$ . Right-skewed distribution.
- **Weibull; Double exponential; Logistic; Pareto ...**

In this section, you should learn

- 1 Random phenomenon behind each distribution and what type of data a distribution is suitable for modeling
- 2 Relationship among distributions
- 3 Meaning of parameters in each distribution
- 4 How to derive cdf/mgf/mean/variance from pdf/pmf ( for most commonly used distributions only)
- 5 **But you are not necessary to memorize** their pdf/pmf/cdf/mgf/mean/variance...



Adapted from Leemis, L. M. (1986), "Relationships among common univariate distributions," *The American Statistician*, Vol. 40, No. 2, pp. 143-146. With permission.

## Types of convergence

i.i.d sample:  $X_1, \dots, X_n$

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

$$P(|\bar{X}_n - \mu| < \varepsilon)$$

random variables  $n \rightarrow \infty$ 

limit  $\bar{X}_n = \mu$   
 $n \rightarrow \infty$   
 $X_1, \dots, X_n$

$\forall \varepsilon > 0, \exists N, \text{ when } n > N$   
 we have  $|\bar{X}_n - \mu| < \varepsilon$

只需要知道 concepts

## Question.

- 1 Toss a fair coin 2 or 3 times. Can you accurately predict the average appearance of heads?
  - 2 Toss a fair coin many many times. What will you predict the average appearance of heads?
- Some *deterministic* patterns emerge from *random* phenomena when more and more data are collected.
  - In the following,  $n \rightarrow \infty$  can be interpreted as **sample size of data is large enough**.

(条件更强)

- **Converge almost surely.** A sequence of random variables  $\{Z_n\}$  is said to **converge almost surely** to a random variable  $Z$ , denoted by  $Z_n \xrightarrow{a.s.} Z$ , if for any  $\varepsilon > 0$ ,

$$P(\lim_{n \rightarrow \infty} |Z_n - Z| < \varepsilon) = 1.$$

$$P(\lim_{n \rightarrow \infty} |Z_n - Z| \geq \varepsilon) = 1 - P(\lim_{n \rightarrow \infty} |Z_n - Z| < \varepsilon) = 0$$

most frequently used

- **Converge in probability.** A sequence of random variables  $\{Z_n\}$  is said to **converge in probability** to a random variable  $Z$ , denoted by  $Z_n \xrightarrow{P} Z$ , if for any  $\varepsilon > 0$ ,

$$\lim_{n \rightarrow \infty} P(|Z_n - Z| < \varepsilon) = 1.$$

- **Converge in distribution.** Let  $Z_1, Z_2, \dots$  be a sequence of random variables with cdf's  $F_1, F_2, \dots$  and let  $Z$  be a random variable with cdf  $F$ . Then  $Z_n$  **converges in distribution** to  $Z$ , denoted by  $Z_n \xrightarrow{d} Z$ , if

$$\lim_{n \rightarrow \infty} F_n(z) = F(z)$$

at every point  $z$ , where  $F$  is continuous.

- **Properties of 3 types of convergence.**

- 1  $Z_n \xrightarrow{a.s.} Z \Rightarrow Z_n \xrightarrow{P} Z$
- 2  $Z_n \xrightarrow{P} Z \Rightarrow Z_n \xrightarrow{d} Z$
- 3  $Z_n \xrightarrow{d} c \Rightarrow Z_n \xrightarrow{P} c$ ,  $c$  is a constant.
- 4 Convergence is preserved by continuous transformation.

## Properties of 3 types of convergence (cont.)

5 Slutsky's theorem: If  $X_n \xrightarrow{d} X$ ,  $Y_n \xrightarrow{P} a$ , where  $a$  is a constant, then

$$(a) Y_n X_n \xrightarrow{d} aX$$

$$(b) Y_n + X_n \xrightarrow{d} X + a$$

$$(c) \frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{a}, \quad P(Y_n \neq 0) = 1 \text{ and } a \neq 0.$$

6 Limit theorem for  $\delta$  method: Suppose that

$$\frac{\sqrt{n}(Y_n - \theta)}{\sigma} \xrightarrow{d} N(0, 1).$$

For a given function  $g$ , suppose that  $g'(\theta) \neq 0$  exists. Then

$$\frac{\sqrt{n}[g(Y_n) - g(\theta)]}{\sigma|g'(\theta)|} \xrightarrow{d} N(0, 1).$$

# Law of Large Numbers

- **Law of large numbers (LLN).** If  $X_1, \dots, X_n, \dots$  be a sequence of independent random variables with  $E(X_i) = \mu$  and  $Var(X_i) = \sigma^2 < \infty$

Let  $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ . Then  $\bar{X}_n \xrightarrow{P} \mu$ . *n large enough,  $\bar{X}_n \approx \mu$*

- **Example 1.20 (Monte Carlo integration).** To calculate  $I(f) = \int_0^1 f(x) dx$ , we can generate  $X_1, \dots, X_n$  i.i.d.  $\sim U(0, 1)$  and compute

$$\int_0^1 \frac{1}{\pi \sqrt{1-x^2}} dx = \int_0^1 f(x) dx$$

*"*  
 $E(g(x)) = \int_0^1 f(x) f_x(x) dx$   
 $f_x(x) = \int_0^1 \frac{1}{\pi \sqrt{1-x^2}} dx$  pdf of  $U(0,1)$

By LLN,  $\hat{I}(f)$  will be close to  $E[f(X)] = \int_0^1 f(x) dx = I(f)$  as  $n$  is large.

*sample  $x_1, \dots, x_n \sim U(0,1) \Rightarrow g(x_1), \dots, g(x_n) \Rightarrow \sum_{i=1}^n g(x_i)/n \approx I$*

$X_i \sim N(\mu, \sigma^2)$   $\bar{X} \sim \text{Normal}$  exactly  
 $X_i \sim F(b)$  i.i.d.  $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \approx \text{Normal}$

# Central Limit Theorem

- Central Limit Theorem.** Let  $X_1, X_2, \dots$  be i.i.d with mean  $\mu$  and variance  $\sigma^2$ . Let  $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$  be the average and  $T_n = n\bar{X}_n$  be the sum of data. Then

$$\lim_{n \rightarrow \infty} P\left(\frac{T_n - n\mu}{\sigma\sqrt{n}} \leq x\right) = \lim_{n \rightarrow \infty} P\left(\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \leq x\right) = \Phi(x),$$

for  $-\infty < x < \infty$ , where  $\Phi(x)$  is the cdf of  $N(0, 1)$ .

$\mu_n = \begin{cases} 1 & p \\ 0 & 1-p \end{cases} \sum X_i$

- Example 1.21** (Normal approximation to Binomial distribution). Let  $X_1, X_2, \dots$  be i.i.d  $\sim B(1, p)$ , then  $T_n \sim B(n, p)$ . Note that  $E(X_i) = p$ ,  $\text{Var}(X_i) = p(1-p)$  and  $E(T_n) = np$ ,  $\text{Var}(T_n) = np(1-p)$ . By CLT,

$$\frac{T_n - np}{\sqrt{np(1-p)}} \xrightarrow{d} N(0, 1),$$

i.e., when  $n$  is large enough, we can approximate the distribution of  $B(n, p)$  by  $N(np, np(1-p))$ .

**Question.** How about those distributions that can be expressed as a sum of other distributions?



### Example 1.22 (Sampling error).

- Suppose that you want to know the average income of families living in Singapore. If you can ask every family for its income, you will get the right answer, denoted by  $\mu$ .
- What if you only take a random sample of , say 1000 families?
- The average income of the 1000 families, denoted by  $\bar{X}_{1000}$ , will have an error  $\bar{X}_{1000} - \mu$ , which is called **sampling error or measurement error**.
- How to assess this error?



**Example 1.23 (Experimental error).** Why many statistical methods assume a normal distributed error term?

- It is usually true that an experimental error  $\varepsilon$  is a function of a number of component errors  $\varepsilon_1, \dots, \varepsilon_n$ . E.g., errors in the settings of experimental conditions, errors due to variation in raw materials, etc.
- If each individual component error is fairly small, it is possible to approximate the overall error as a linear function of independently distributed component errors  $\varepsilon = a_1\varepsilon_1 + \dots + a_n\varepsilon_n$ .
- By CLT,  $\varepsilon$  will tend to be normal as the number of components becomes large.